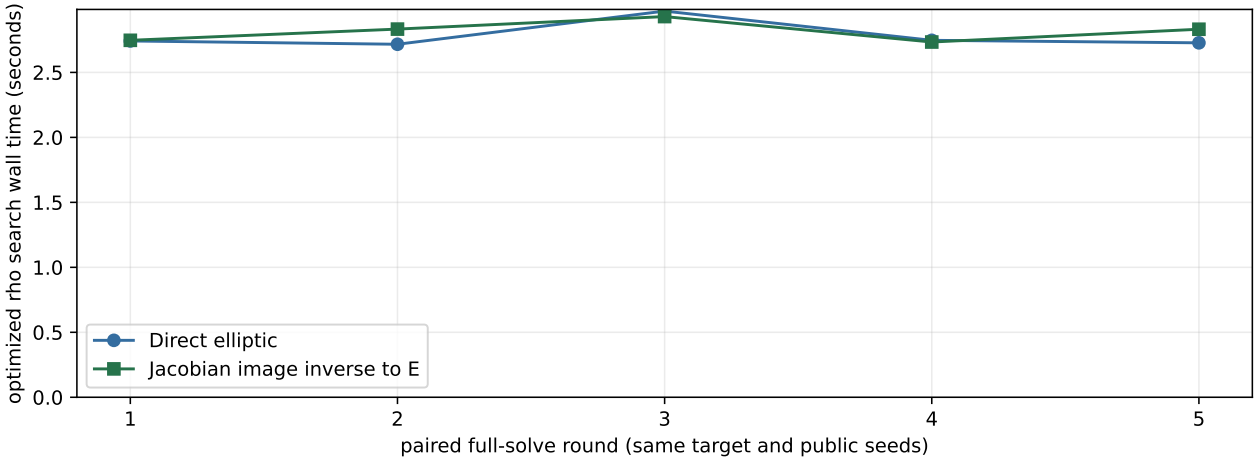
Full-subgroup ECDLP on an elliptic curve and a genus-two image

GF(2^{83}), $a = 0, 1$; exploratory current-worker campaign, 14 October 2026

The source is $E_a : y^2 + xy = x^3 + ax^2 + 1$. The constructed curve $C_a : v^2 + u^2 v = u^5 + au^4 + u^3 + u$ has genus two, and its degree-three cover induces an injective pullback on the pinned prime-order subgroups. The comparison concerns this elliptic image inside $J(C_a)$. Its group law is Jacobian addition; the image also has a checked inverse to E_a.

Exact workload. Polynomial basis $z^{83} + z^{45} + z^2 + z + 1$; source identities `EC1N83Ckb0h252dacc102a4` and `EC1N83Ckb1h1f579d351b52`. Prime orders are $r_0 = 241785163923079621668568 9$ and $r_1 = 8569786107849059$. The latter is a 53-bit subgroup, despite the 83-bit field. Pinned generators, public coordinates, subgroup cofactors and job hashes are in the frozen public-job files. Eight nonzero targets per curve were drawn uniformly using OS rejection sampling. Scalar witnesses were written outside solver inputs; no solver command accepts a witness or an interval.

Completed optimized runs. The reference suite performs 27 full $a = 1$ runs across eight independent targets: one warmup, two A/A controls, five paired direct/image-inverse rounds on target zero, and a direct/image pair on each of the seven other targets. Every recovered candidate is verified in another process, by both elliptic and native Jacobian scalar multiplication. Repeating a target does not increase the independent-target count.

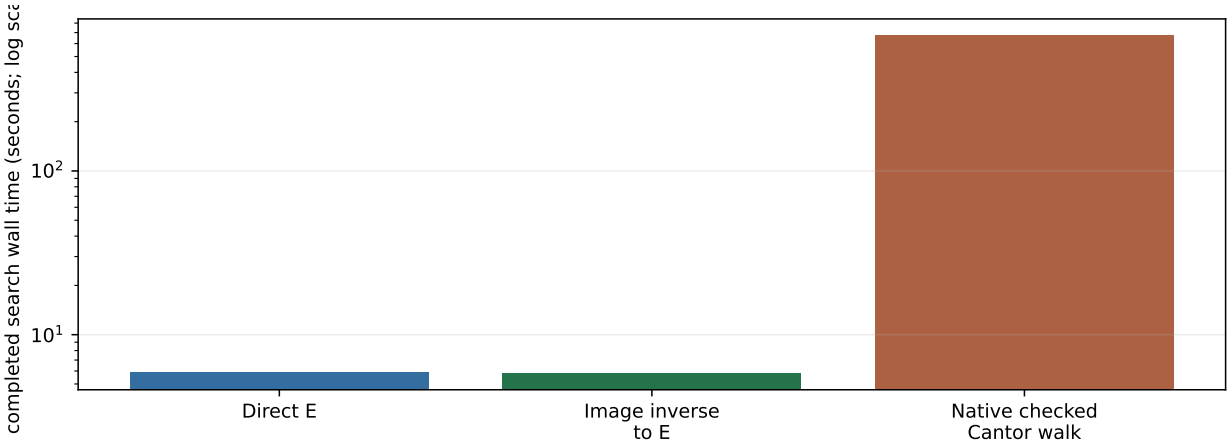


a=1 target 0; every run recovered and externally verified the full-order log. L0; no admitted speed ratio.

The image-inverse arm constructs the two public divisors, checks their inverse, and runs the same optimized rho on E_a. Transfer, setup, search and verification are recorded separately; the graph shows search only. This establishes a working full-order transfer-and-solve route. It does not establish an ECDLP advantage from changing the representation.

Native Jacobian search. The separately named resumable method uses 32 lockstep lanes and 32 public jumps, distinguished points, all 166 signed Frobenius actions, short-cycle escape and a persistent collision table. All three arms follow the same matched walk. The elliptic arm uses individual additions; it is distinct from the optimized batched reference on page one. The Jacobian arm uses checked Cantor addition, then a checked image inverse and pullback to choose canonical representatives. These conversions and repeated backend validation are included in its cost. This is a conservative software implementation, not a benchmark of optimized genus-two formulas or index calculus.

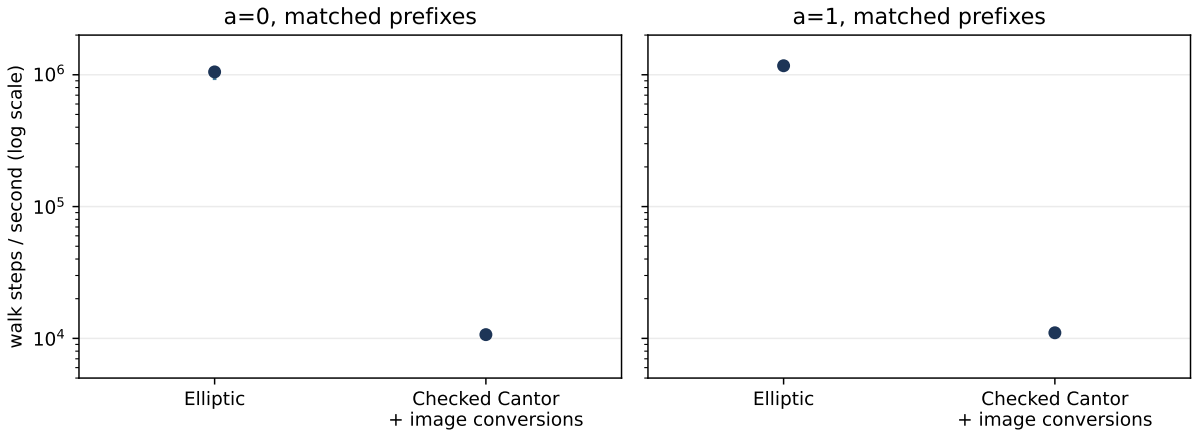
Matched resumable arm	Steps	Search seconds	Status
Direct E	6,631,641	5.846	verified
Image inverse to E	6,631,641	5.842	verified
Native checked Cantor	6,631,641	669.526	verified



One full-subgroup solve per arm, a=1 target 0; same resumable method. L0 descriptive timings.

Rate calibration. One warmup, two A/A controls and five interleaved E/Cantor prefix pairs were recorded per curve. Matching lane digests and restarted/uninterrupted state checks pass in both repositories. Prefixes are not completed solves. Counts of E additions and Cantor additions are distinct native work units; their ratio is not a priced cost ratio.

a	Arm	Median steps/s	Observed range
0	elliptic	1,051,224	911,331–1,101,530
0	checked Cantor	10,678	10,562–11,008
1	elliptic	1,170,906	1,153,715–1,175,601
1	checked Cantor	11,030	10,387–11,158



Five A/B rounds: median and observed range; L0 VM timings. Prefixes are not full solves.

Host and admissibility. Intel Xeon Platinum 8573C virtual worker, five exposed CPUs, four-CPU quota and 16 GiB memory limit. The native ecbench primitives take the exclusive benchmark lock, attempt eviction and pin the solver to CPU 4. Actual affinity is recorded as CPU 4; the observed memory policy is default. Physical memory type is unavailable. The preflight could not read CPU/memory PSI, some user threads could not be moved, and hardware instruction/cycle counters returned `perf_event_open: No such file or directory`. These are L0 descriptive VM timings. No admitted speedup, common priced-work ratio, independent-host performance replay or sealed ecbench audit is claimed. No ARM, GPU or FPGA run was performed.

Long campaign and persistence. The checkpointed worker searches the complete prime subgroups for both a values. It serially schedules six streams and advances each arm through eight targets after independent verification. The $a = 1$ native Jacobian stream receives longer quanta to obtain a full native comparison; $a = 0$ work is preserved. Live progress is distinct from frozen snapshot evidence. Neither an unfinished search nor a birthday-scale rate projection is a completed solve duration. Current-worker jobs are not durable against deletion of that worker; the raw checkpoints and exact resume commands must be retained or moved to another machine. For $a = 0$, the ideal birthday scale divided by the native prefix rate is about 1.42×10^7 seconds on a fully occupied core, before scheduling overhead; this months-scale heuristic is not a measured duration or calibrated ETA.

Checkpoints include every stored distinguished point, coefficients, lane histories, cursor and seed/input identity. Writes are atomic and synced. A pause marker causes a complete snapshot before other heavy work; a stop marker ends orderly. Reconstruction of native divisor caches is timed as resume work. Elapsed time determines scheduling only, never a scalar-space restriction.

Validation and remaining gates. Both packages pass 33 native unit tests plus three cover tests, formatting, scoped Clippy, CLI rejection controls and exact map-identity replay. Identical public-job restart receipts were obtained in both repositories. The constructed collision unit test is a correctness control, not a benchmark solve. Backend source remains unchanged; the preceding whole-library gate still has its two documented JV F4 completion threshold failures. This report does not call that gate green.

The existing ecbench wide constructor assumes cofactor 4 or 2 and rejects the $a = 1$ pinned prime-factor subgroup. Its counted-group workload has no Mumford image adapter. The separate native campaign therefore has no ecbench audit certificate. A registered width-generic adapter, common native-work calibration, admissible isolation and independent-host replay remain open requirements. The earlier cover report's geometric Lean bridge also remains partial.

Evidence and sources. Preregistered protocol:

`research/binary_genus2_speed_20261014/PROTOCOL.md`

Frozen `runs/run-20261014T1901Z/`, including public jobs, raw per-round/reference records, verification receipts, host manifests, checkpoint snapshot, source and binary bindings and SHA-256 manifests. The native summary supplies every plotted value; the renderer only lays out these recorded values.

Construction proof and identity certificate: binary genus-two construction study. Campaign source and invocation: native Rust tool. Remaining harness gate: `problems/ecbench-binary-genus2-full-subgroup-adapter.md`. Canonical IC scoreboards and progress graphs were checked for applicability: this study adds no IC/rho ratio, registered curve or admitted gain, so those series remain unchanged. The performance visual index receives this report.